



Batterjee Medical College
For Science and Technology

Approach to a case of Hypertension

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Hypertension

Learning Objectives

Knowledge & understanding

Define hypertension and classify blood pressure categories according to current ACC/AHA and JNC guidelines.

Differentiate primary from secondary hypertension based on their underlying pathophysiologic mechanisms.

Skill

Recognize clinical clues suggestive of secondary hypertension to guide a targeted diagnostic workup.

Select an appropriate initial antihypertensive regimen based on blood pressure severity, comorbidities, and patient-specific factors.



Main Subtopics

Definitions and Classification of Blood Pressure

Types and Subtypes

Diagnostics and Screening

Treatment

- Lifestyle modification
- Indications for pharmacotherapy
- Initial medication selection
- First-line and second-line agents
- Long-term Management
- Follow-up

Complications

Hypertensive Crises

Hypertensive Disorders of Pregnancy

Case scenario :

A previously healthy 10-year-old boy is brought to the emergency department for the evaluation of one episode of vomiting and severe headache since this morning. His mother says he also had difficulty getting dressed on his own. He has not had any trauma. The patient appears nervous. His temperature is 37.0 °C (98.6°F), pulse is 100/min, and blood pressure is 185/125 mm Hg. He is confused and oriented only to person. Ophthalmic examination shows bilateral optic disc swelling. There is an abdominal bruit that is best heard at the right costovertebral angle. A complete blood count is within normal limits.

Case problem :

Which of the following is most likely to confirm the diagnosis?

- A. Oral sodium loading test
- B. CT angiography
- C. Serum IGF-I level
- D. Serum PTH level
- E. High-dose dexamethasone suppression test
- F. Serum 17-hydroxyprogesterone level
- G. Urinary catecholamine metabolites

Hypertension

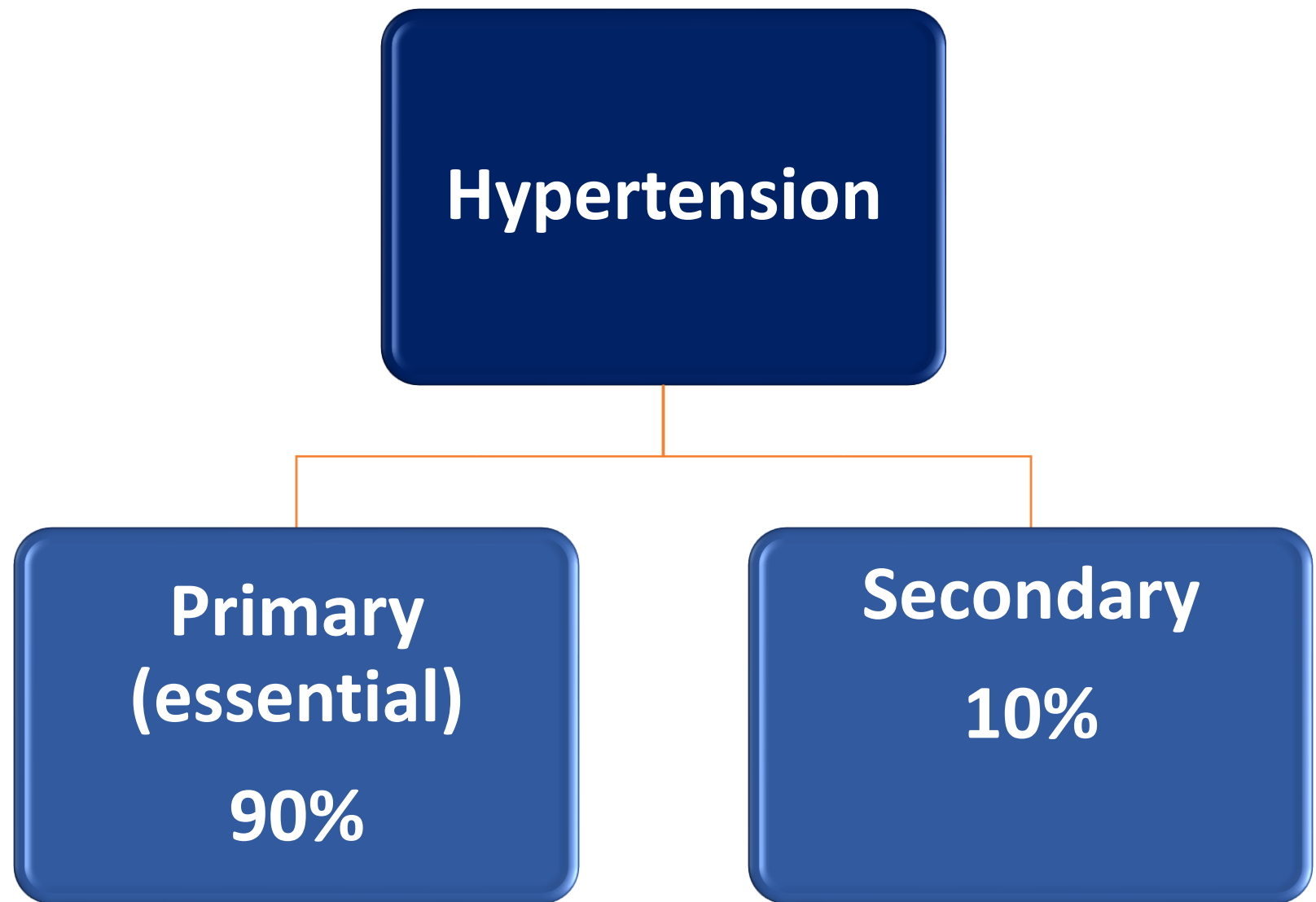
The 2017 and 2022 American College of Cardiology (ACC)/American Heart Association (AHA) guidelines define hypertension in adults as a blood pressure of $\geq 130/80$ mm Hg.

The Eighth Joint National Committee (JNC 8) criteria specify $\geq 140/90$ mm Hg.

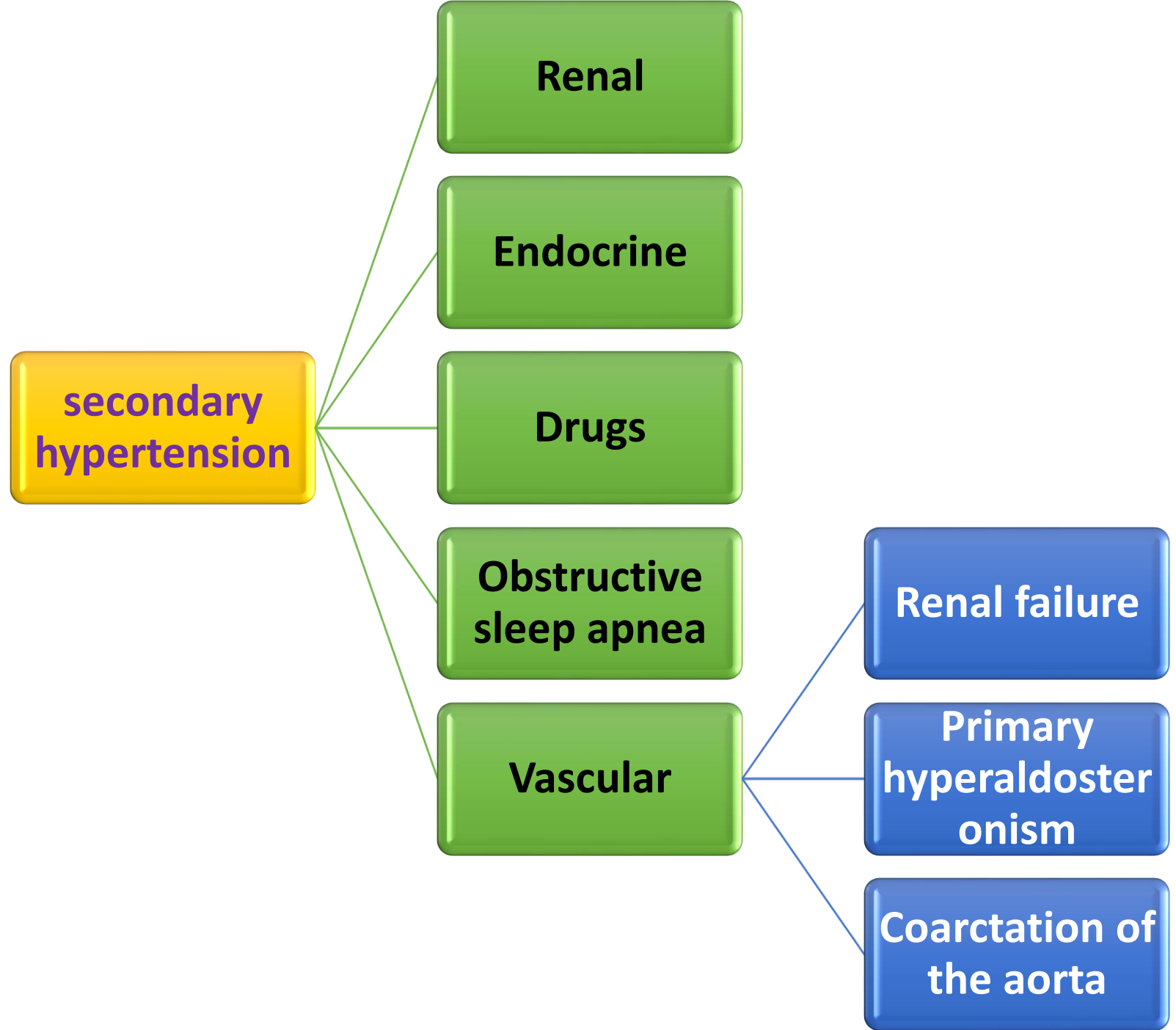
Resistant hypertension: hypertension that remains uncontrolled ($\geq 130/80$ mm Hg) despite treatment with ≥ 3 antihypertensives OR requires ≥ 4 medications to be controlled

A common condition that affects one in every three adults.

Type of Hypertension



Underlying conditions of secondary hypertension



Etiology

Primary hypertension

Directly related to total peripheral resistance and cardiac output

Risk factors for primary hypertension

Non-modifiable risk factors

- Positive family history
- Ethnicity
- Advanced age

Modifiable risk factors

- Obesity (greatest modifiable risk factor)
- Diabetes
- Smoking
- Excessive alcohol intake
- Diet high in sodium
- Physical inactivity
- Psychological stress

Blood Pressure Categories



BLOOD PRESSURE CATEGORY	SYSTOLIC mm Hg (upper number)		DIASTOLIC mm Hg (lower number)
NORMAL	LESS THAN 120	and	LESS THAN 80
ELEVATED	120 – 129	and	LESS THAN 80
HIGH BLOOD PRESSURE (HYPERTENSION) STAGE 1	130 – 139	or	80 – 89
HIGH BLOOD PRESSURE (HYPERTENSION) STAGE 2	140 OR HIGHER	or	90 OR HIGHER
HYPERTENSIVE CRISIS (consult your doctor immediately)	HIGHER THAN 180	and/or	HIGHER THAN 120

Comparison of Classifications of Blood Pressure in ACC/AHA 2017, ESH 2023, and ESC 2024 Guidelines

Systolic BP (mmHg)	120	130	140	150	160	170	180
ACC/AHA 2017	Normal	Elevated	Stage 1	Stage 2 Hypertension			
ESH 2023	Optimal	Normal	High-normal	Grade 1 Hypertension	Grade 2 Hypertension		Grade 3 Hypertension
ESC 2024	Non-elevated	Elevated		Hypertension			

Diastolic BP (mmHg)							
ACC/AHA 2017	Normal		Stage 1 Hypertension		Stage 2 Hypertension		
ESH 2023	Optimal		Normal	High-normal	Grade 1 Hypertension	Grade 2 Hypertension	Grade 3 Hypertension
ESC 2024	Non-elevated	Elevated			Hypertension		
	70	80	90	100	110		

Definitions and Classification of BP (AHA-2025)

2.1. Definition of High BP

Recommendation for Definition of High BP

References that support the recommendation are summarized in the [Evidence Table](#).

COR	LOE	Recommendation
1	B-NR	1. In adults, BP should be categorized as normal, elevated, or stage 1 or 2 hypertension to prevent and treat high BP (Table 4). ^{1,2}

Table 4. Categories of Blood Pressure in Adults* ([Table view](#))

	SBP		DBP
BP Category			
Normal	<120 mm Hg	and	<80 mm Hg
Elevated	120 to 129 mm Hg	and	<80 mm Hg
Hypertension			
Stage 1	130 to 139 mm Hg	or	80 to 89 mm Hg
Stage 2	≥140 mm Hg	or	≥90 mm Hg

BP indicates blood pressure (based on an average of ≥2 careful readings obtained on ≥2 occasions, as detailed in [Section 3](#) ("Evaluation and Diagnosis"); DBP, diastolic blood pressure; and SBP, systolic blood pressure.

* Adults with SBP and DBP in 2 categories should be designated to the higher BP category. This table excludes individuals who are pregnant (see [Section 11.5](#), "Hypertension and Pregnancy"). Adapted with permission from Whelton et al.⁵ Copyright 2018 American College of Cardiology Foundation and American Heart Association, Inc.

Blood Pressure Thresholds and Recommendations for Treatment and Follow-Up

LEGEND

COR 1

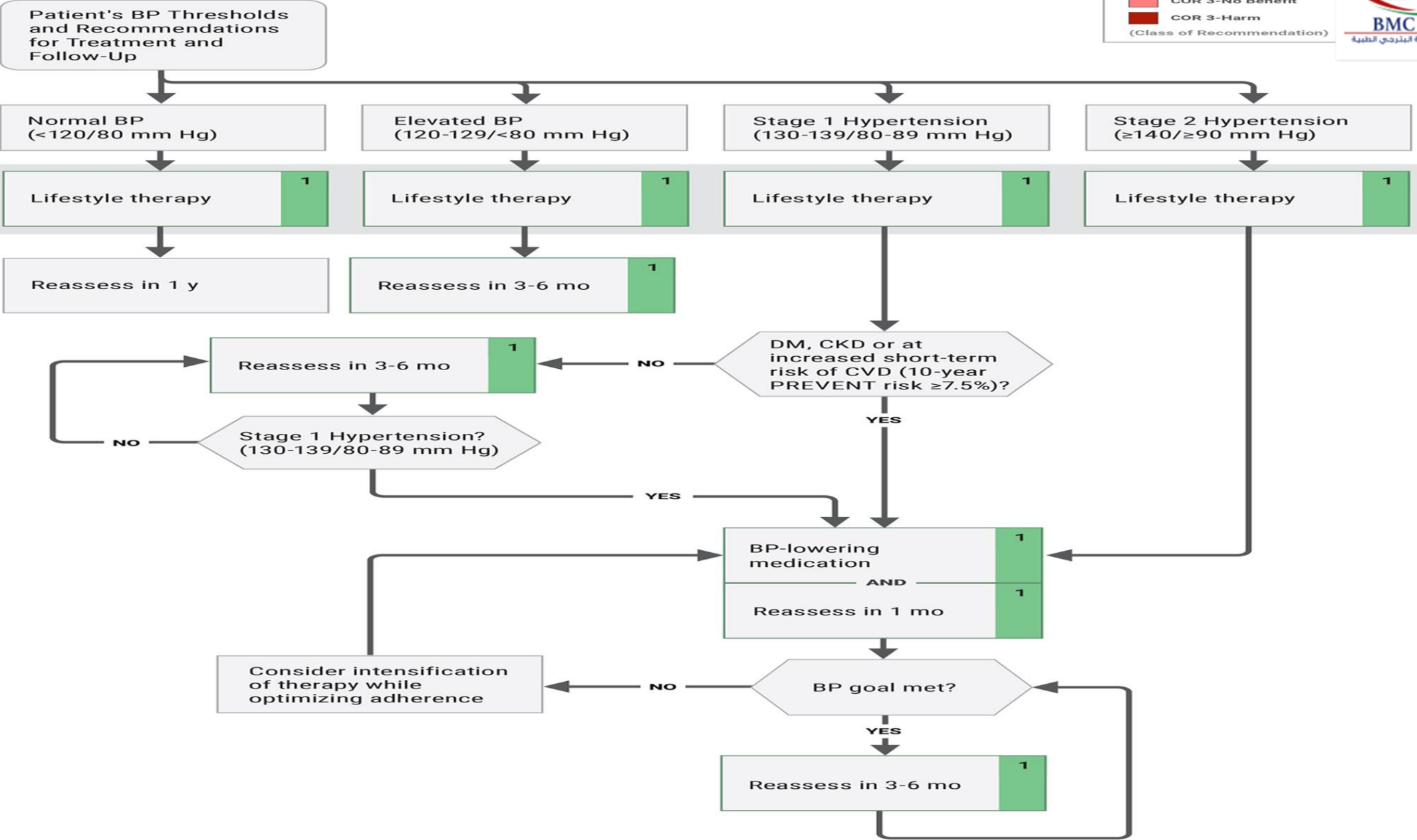
COR 2a

COR 2b

COR 3-No Benefit

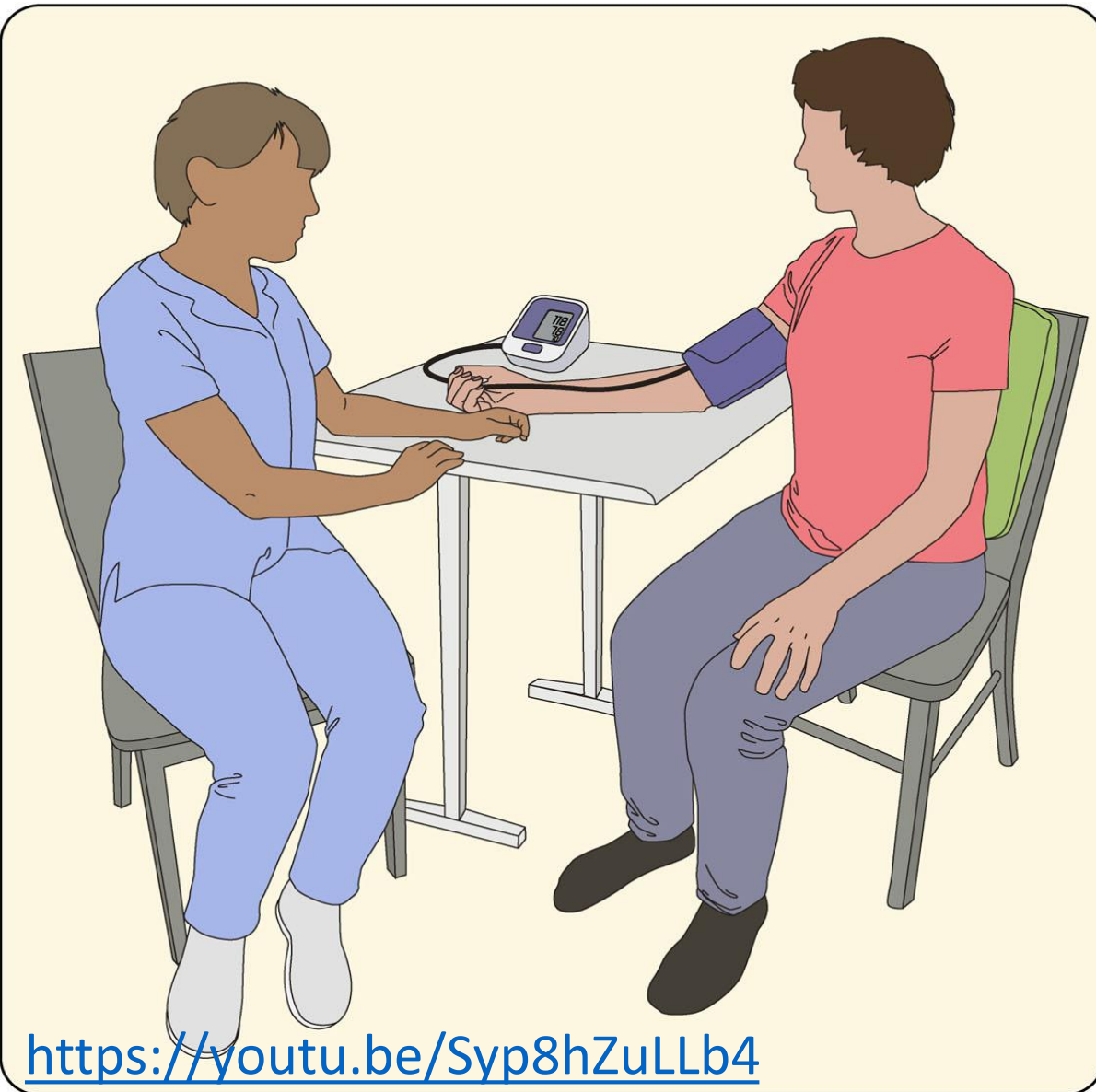
COR 3-Harm

(Class of Recommendation)



Blood Pressure Measurement

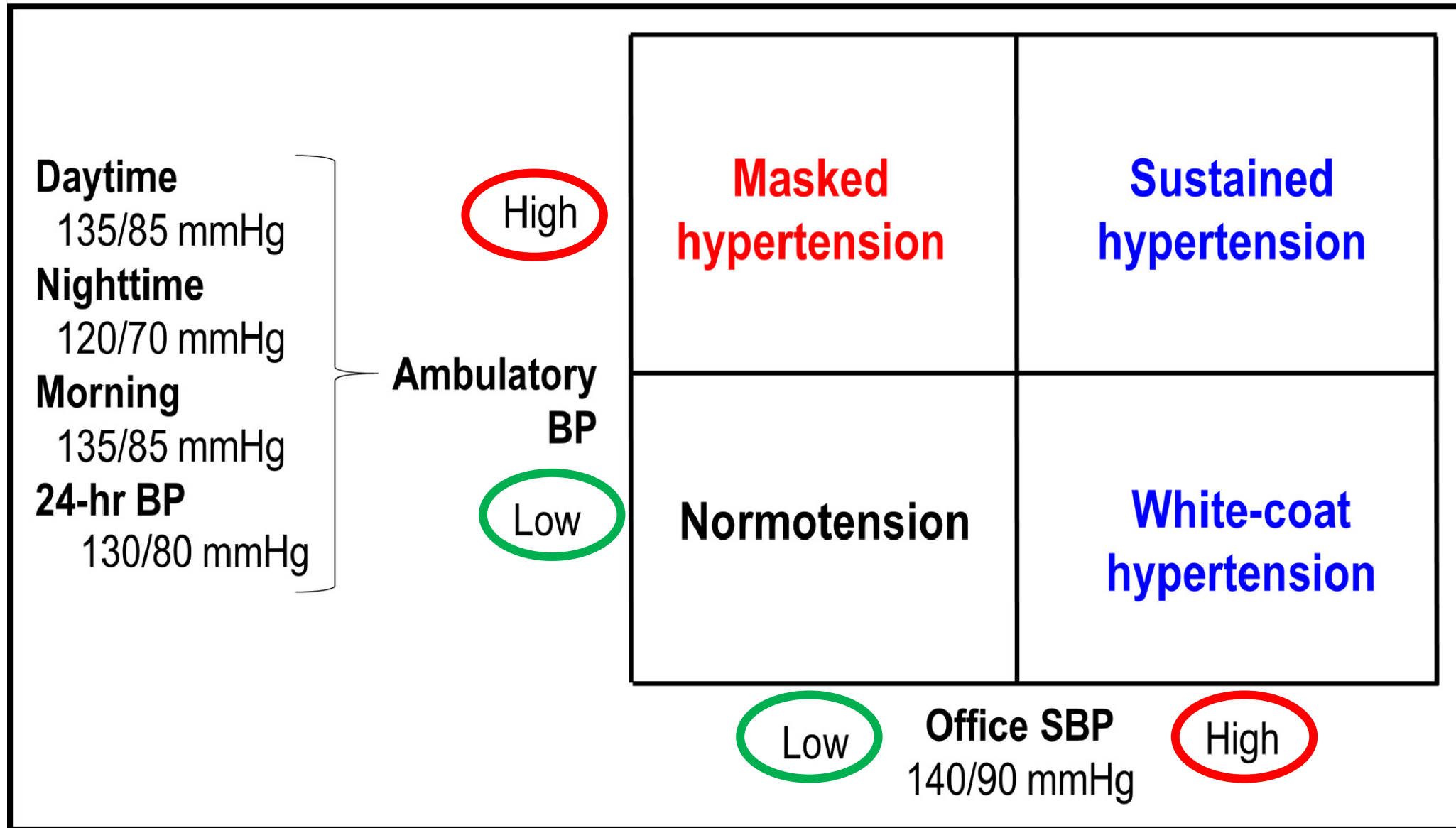
Office Blood Pressure Measurement



1. The patient should avoid caffeine, exercise, and smoking for at least 30 minutes before measurement. Ensure the patient has emptied their bladder.
2. Use a blood pressure device that has been validated for accuracy (validatebp.org).
3. Use the correct cuff size on a bare arm.
4. The patient's arm should be supported at heart level.
5. Have the patient relax, sitting in a chair (feet on floor, legs uncrossed, and back supported) for more than 5 minutes of rest.
6. Neither the patient nor the clinician should talk during the rest period or during the measurement. The patient should not be using their phone.
7. Blood pressure measurement should be taken in a temperature-controlled room.
8. Take 2 or more blood pressure measurements at least 1 minute apart. Average the readings, and provide the patient their blood pressure readings both verbally and in writing.

<https://youtu.be/Syp8hZuLLb4>

Subtypes and variants



Isolated systolic hypertension

Definition

Elevated SBP (≥ 140 mm Hg) with DBP within normal limits (≤ 90 mm Hg)

Etiology

Decreased arterial elasticity and compliance due to aging

Treatment

- Recommend lifestyle changes for managing hypertension.
- Start pharmacological treatment of hypertension.

Thiazide diuretics **or** dihydropyridine calcium antagonists

Clinical features

Hypertension is usually **asymptomatic** until:

- Complications of end-organ damage arise
- Or an acute increase in blood pressure occurs “Hypertensive crisis”

Secondary hypertension usually manifests with symptoms of the underlying disease.

Hypertensive crises

Acute blood pressure increase
 ≥ 180 mm Hg systolic and/or ≥ 120 mm Hg diastolic

Hypertensive urgency

Hypertensive crisis **without** signs of acute organ damage

Headache

Lightheadedness, vertigo

Epistaxis



Restlessness, insomnia

Hypertensive emergency +

Hypertensive crisis **with** signs of acute organ damage

Intracranial hemorrhage, stroke

Hypertensive encephalopathy

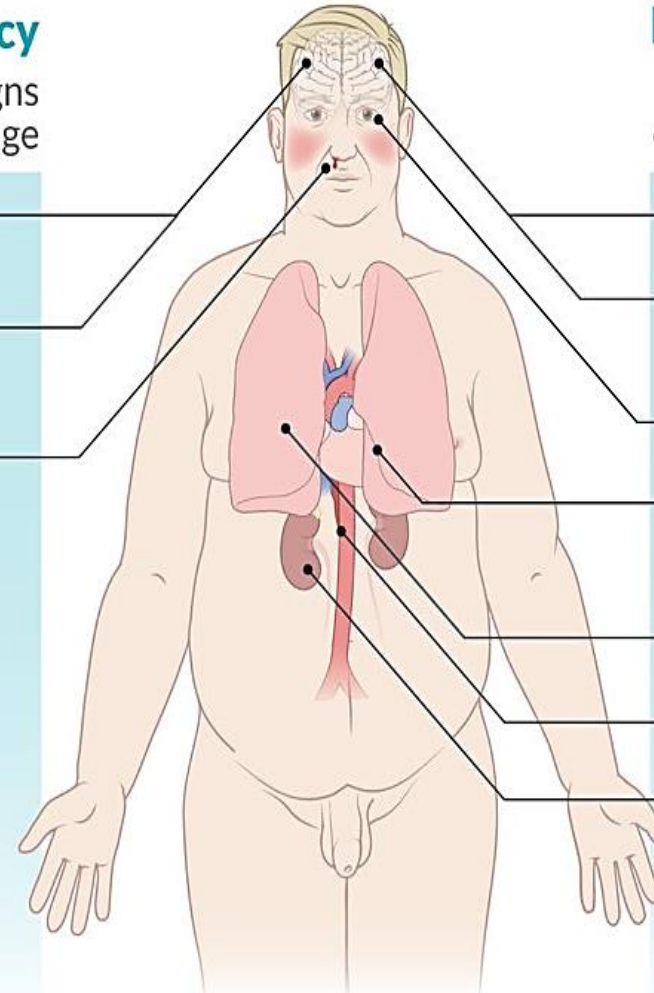
Retinal hemorrhages

Hypertensive acute heart failure, myocardial infarction

Hypertensive pulmonary edema

Aortic dissection

Renal failure



Causes of secondary hypertension

Causes of secondary hypertension

- Most common causes in adults include:

RECENT:

Renal

Endocrine

Coarctation of the aorta

Estrogen (oral contraceptives)

Neurological (raised intracranial pressure, psychostimulants use)

Treatment (Drugs)

Causes of secondary hypertension

Signs suggestive of secondary hypertension

- Unusual onset of hypertension

Onset at < 30 years of age

- **Aortic dissection is a (rare)**
life-threatening sign of secondary hypertension that may manifest with a **blood pressure difference between the right and left arm.**
- **Coarctation of the aorta**
Secondary hypertension that may manifest with a **blood pressure difference between the upper and lower limb.**

Renal hypertension

1. Renal artery stenosis (e.g., due to atherosclerosis, fibromuscular dysplasia, polyarteritis nodosa)

Potential indications for further workup

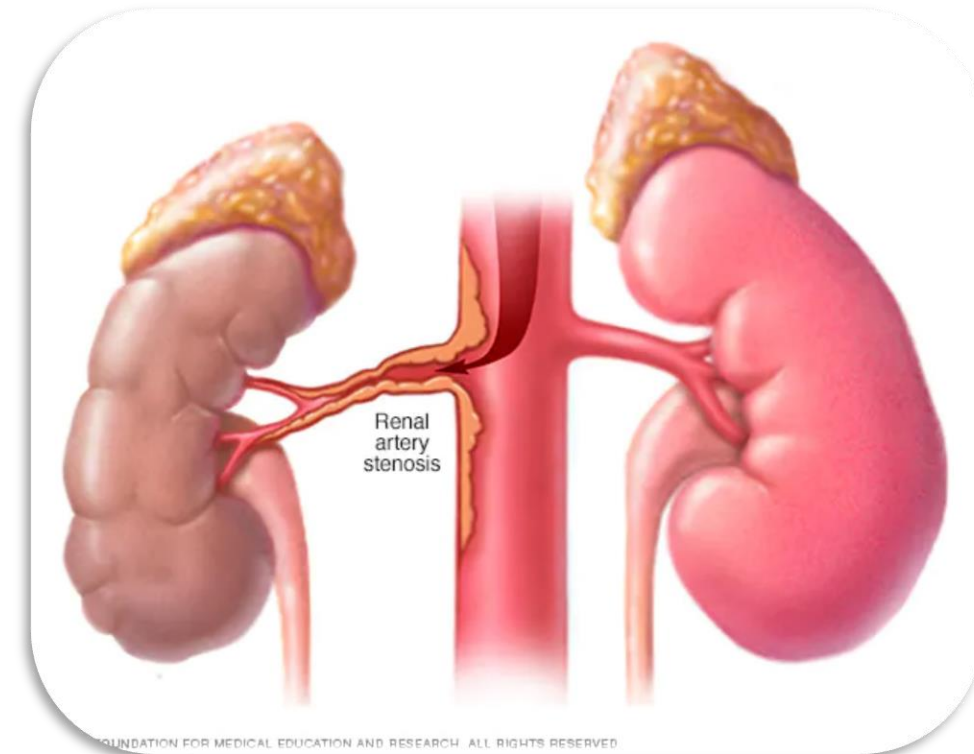
- Recurrent flash pulmonary edema
- Abdominal bruit
- $\uparrow$ Serum creatinine (by $\geq 50\%$)
within 1 week of starting an ACEI or ARB
- Hypokalemia
- Asymmetric kidney size

2. Renal parenchymal disease:

Glomerulonephritis

Polycystic kidney disease

3. Chronic kidney disease



Common causes of endocrine hypertension:

- **Primary hyperaldosteronism (Conn syndrome)**
- **Pheochromocytoma**
- **Hypercortisolism (Cushing syndrome)**
- **Hyperthyroidism**
- **Primary hyperparathyroidism**
- **Congenital adrenal hyperplasia**
- **Acromegaly**

Pheochromocytoma

- Resistant hypertension
- Paroxysmal hypertension
- Episodes of headaches
- Palpitations and
- Diaphoresis

"5 P's"

SIGNS AND SYMPTOMS OF PHEOCHROMOCYTOMA

*PRESSURE (HYPERTENSION)

*PAIN (HEADACHE)

*PERSPIRATION

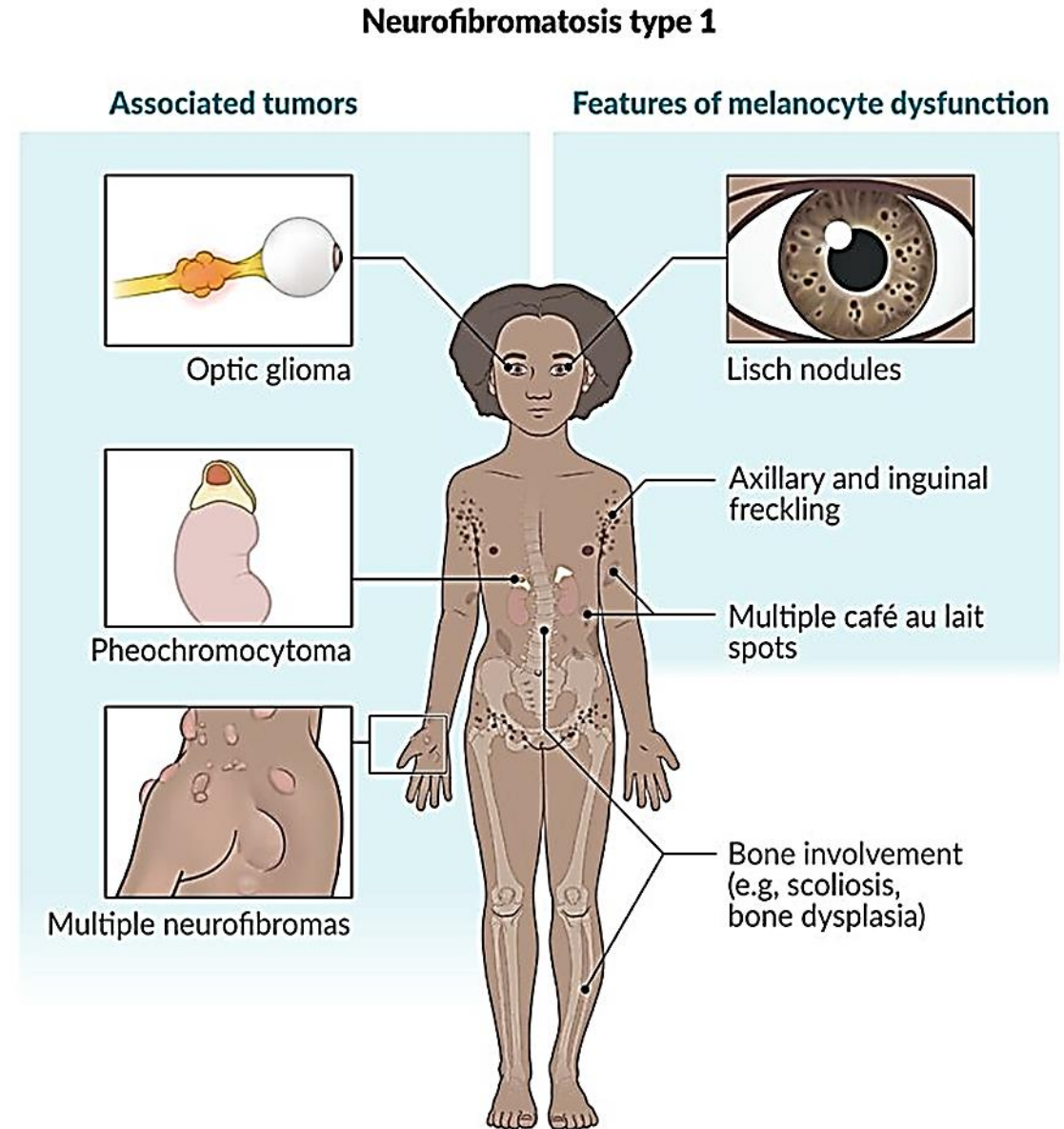
*PALPITATION

*PALLOR



Pheochromocytoma

- Family history of endocrine tumors
 - Cutaneous features suggestive of NF type 1
- ↑ 24-hour urinary fractionated metanephrines
- ↑ Plasma metanephrines



Causes of secondary hypertension

Coarctation of the aorta distal to the left subclavian artery

Blood pressure difference between the upper and lower limbs

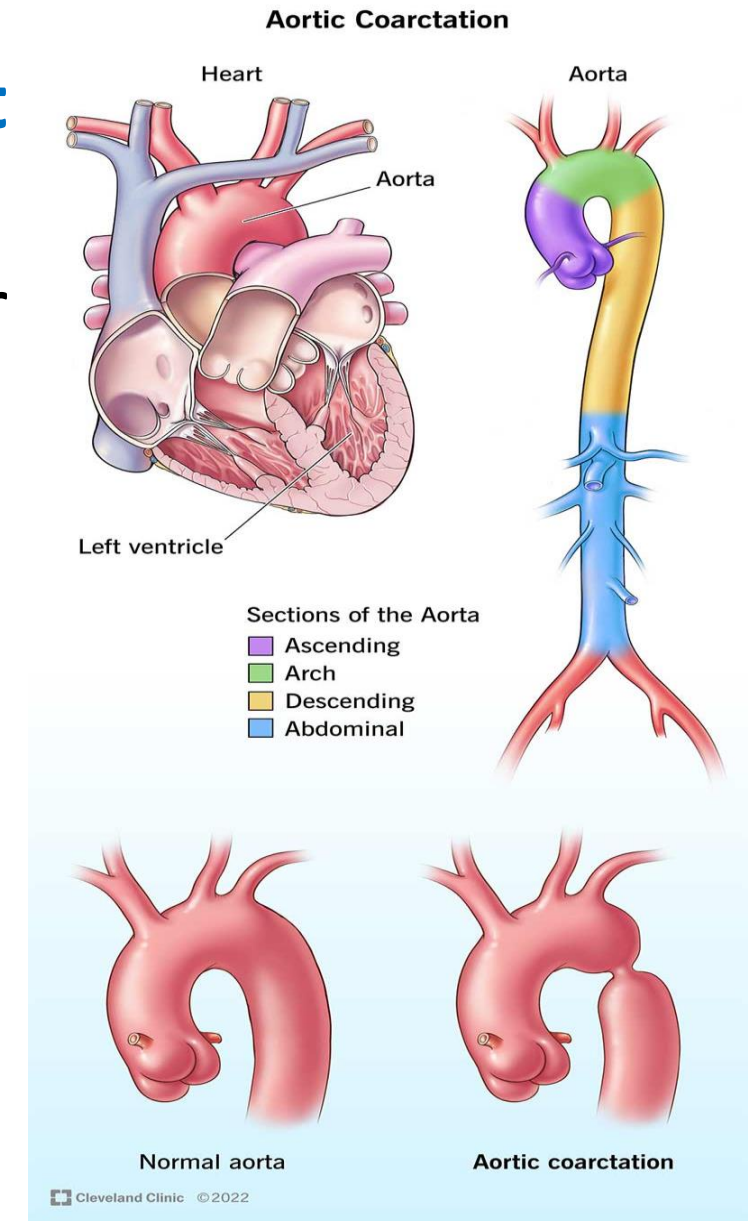
Obstructive sleep apnea

Potential indications for further workup

Resistant hypertension

daytime sleepiness

➤ Sleep studies often leads to resolution of hypertension.



Substance-related

Recreational drug use:

- Amphetamines
- Cocaine
- Phencyclidine

Use of certain medications:

- Sympathomimetic drugs
- Corticosteroids
- NSAIDs
- Oral contraceptives

➤ Urine drug screening

Approach

Consider diagnostic workup for secondary causes of hypertension in patients with signs suggestive of secondary hypertension

Screening for hypertension

Indications

- Annual screening
 - Individuals > 40 years of age
 - Adults of any age with risk factors for primary hypertension
- Screening every 3–5 years:
Individuals 18–39 years of age with previously normal blood pressure (< 130/85 mm Hg) and no risk factors

Evaluation of patients with newly diagnosed hypertension

The initial exam should focus on:

- ✓ Signs indicating secondary hypertension
- ✓ Signs indicating target organ damage
- ✓ The assessment of ASCVD risk.

Routine studies

- Fasting blood glucose
- Serum sodium, potassium, and calcium levels
- Renal function tests: serum creatinine and eGFR
- Lipid profile (HDL, LDL, and triglycerides levels)
- Urinalysis
- Electrocardiogram (ECG)

Additional studies

- Hemoglobin A1c

Approach:

- **Lifestyle changes** for managing hypertension: for all patients with SBP > 120 mm Hg or DBP > 80 mm Hg
- **Titrate** treatment **to** reach **target** blood pressure.;
Most adults: blood pressure < 130/80 mm Hg
- **Follow-up regularly**: reassess indications for pharmacological treatment and tailor therapy to individual needs

Use of Risk-Based Thresholds for Initiation of BP Treatment in Adults

BP Level-Only

Does the patient have an average BP $\geq 140/90$ mm Hg?

YES

Initiate antihypertensive medications to lower BP and reduce CVD risk for primary or secondary prevention of CVD

1

NO

Risk-Based Thresholds for Initiation of BP Treatment for Adults*

Does the patient have existing clinical CVD (CHD, stroke, HF)?

YES

Initiate antihypertensive medications to lower BP and reduce CVD risk if average SBP ≥ 130 mm Hg or DBP ≥ 80 mm Hg for secondary prevention of CVD

1

NO

Does the patient have diabetes or CKD, or is the patient at increased short-term risk of CVD (10-year PREVENT-CVD risk $\geq 7.5\%$)?

YES

Initiate antihypertensive medications to lower BP and reduce CVD risk if average SBP ≥ 130 mm Hg or DBP ≥ 80 mm Hg for primary prevention of CVD

1

NO

Initiate antihypertensive medications to lower BP if average SBP ≥ 130 mm Hg or DBP ≥ 80 mm Hg after 3-6 months of lifestyle intervention

1

LEGEND

- COR 1
 - COR 2a
 - COR 2b
 - COR 3-No Benefit
 - COR 3-Harm
- (Class of Recommendation)



Treatment

Lifestyle changes for managing hypertension ^{[1][5][19]}

Intervention (in order of effectiveness)		Target	Approximate SBP reduction in hypertensive patients
Weight loss (most effective measure)		• <u>Ideal body weight</u>	• 1 mm Hg per kg reduction in body weight in <u>overweight</u> individuals
Diet	DASH diet	• Diet rich in fruits, vegetables, and whole grains ^[28] • Low in saturated and trans fats	• 11 mm Hg
	Decrease dietary sodium	• Daily <u>sodium</u> intake < 1500 mg ^{[5][25][28]}	• 5–6 mm Hg
	Increase dietary <u>potassium</u>	• Daily <u>potassium</u> intake 3.5–5 g (preferably by increasing fruit and vegetable intake) ^[29]	• 4–5 mm Hg

Effect of regular exercise

Exercise	Aerobic [30]	• 90-150 minutes per week [5][28]	• 5-8 mm Hg
	Dynamic resistance (e.g., weight training)	• 90-150 minutes per week	• 4 mm Hg
	Isometric resistance (e.g., hand grip exercise)	• Three sessions per week	• 5 mm Hg



Pharmacological treatment

Indications for antihypertensive treatment

- SBP ≥ 140 mm Hg or DBP ≥ 90 mm Hg
- Adults with SBP ≥ 130 mm Hg or DBP ≥ 80 mm Hg and ≥ 1 of the following:
 - Clinical ASCVD (e.g., ischemic heart disease, peripheral artery disease, or previous stroke) or congestive heart failure
 - 10-year ASCVD risk $\geq 10\%$; age ≥ 65 years and diabetes mellitus

Initial medication

Choice of initial medication should be based on the following:

- Patient's initial blood pressure;
 1. SBP 130–139 mm Hg or DBP 80–89 mm Hg (**stage 1 hypertension**): Consider initial monotherapy.
 2. SBP $\geq$ 140 mm Hg or DBP $\geq$ 90 mm Hg **And** an average blood pressure $>$ 20/10 mm Hg above target
 - Initiate combination therapy.
 - Commonly used combinations are an ACEI or ARB PLUS either a dihydropyridine CCB OR a thiazide-type diuretic.
- Additional factors to consider
 1. Major comorbidities
 2. For Black patients (including individuals with diabetes) without CHF or CKD, initial antihypertensive therapy should include a thiazide-type diuretic or CCB.

Treatment-First-line options

First-line antihypertensive medications ^{[1][19][25][35]}

Drug class	Examples	Indications
ACEIs	• <u>Lisinopril</u> • <u>Enalapril</u> • <u>Captopril</u>	• Nephroprotective, therefore preferred in patients with: ◦ <u>Diabetes mellitus</u> and <u>albuminuria</u> ◦ <u>Renal disease</u> ^[36]
ARBs	• <u>Losartan</u> • <u>Valsartan</u>	
Thiazide diuretics	• <u>Chlorthalidone</u> • <u>Hydrochlorothiazide</u>	• Preferred: ◦ As one of the initial antihypertensive medications in Black patients ^[2] ◦ In patients with <u>isolated systolic hypertension</u> ^{[17][18]}
Dihydropyridine CCBs ^[19]	• <u>Amlodipine</u> (preferred) • <u>Nifedipine</u>	
Nondihydropyridine CCBs	• <u>Diltiazem</u> • <u>Verapamil</u>	• Less commonly used

Treatment- Second-line options

Beta blockers

Combined alpha- and beta-adrenergic receptor antagonists

Carvedilol - Labetalol

Noncardioselective

Nadolol - Propranolol

Cardioselective

Atenolol – Bisoprolol - Metoprolol succinate

Recommended for first-line use only in patients with the following comorbidities:

- Ischemic heart disease
- Heart failure
- Atrial fibrillation
- Thoracic aortic disease (e.g., aortic dissection, thoracic aortic aneurysm)
- Thyrotoxicosis
- Migraine
- Essential tremor

Treatment- Second-line options

Loop diuretics

Furosemide

Torsemide

Preferred with:

- Symptomatic heart failure
- CKD (if GFR < 30 mL/min)

Side effects of furosemide & thiazides	
4 hypo	4 hyper (glucose)
1. -Hypokalemia (↑ risk of digitalis toxicity)	5. -hyperglycemia
2. -Hypovolemia	6. -hyperlipidemia
3. -Hyponatremia	7. -hyperurecemia
4. -Hypochloremic metabolic alkalosis.	8. -hypercalcemia➡ (Thiazide only)

Potassium-sparing diuretics

Aldosterone antagonists

- **Steroidal** Spironolactone Eplerenone **Nonsteroidal**: finerenone

Preferred in hypertension due to primary hyperaldosteronism

Frequently used as add-on therapy in resistant hypertension

- **Epithelial sodium channel blockers**

Amiloride

Triamterene

Consider as add-on therapy for patients with hypokalemia who are receiving thiazides.

Treatment- Second-line options

Alpha-1 blockers

Prazosin - Doxazosin

Used in hypertension caused by pheochromocytoma

May be used as an adjunct in patients with benign prostatic hypertrophy

Direct arteriolar vasodilators

Hydralazine -Minoxidil

Hydralazine is a treatment option in pregnant patients.

Combined alpha- and beta-adrenergic receptor antagonists

Carvedilol - Labetalol

General principles

Most patients initiated on pharmacological treatment:

- Follow up after ~ 1 month.
- Reassess after 3–6 months and annually thereafter

Medication titration

Adjust medication to reach optimal blood pressure control.

If the treatment goal cannot be reached with two drugs:

- ✓ Add a third drug.
- ✓ Evaluate for secondary hypertension, if indicated.

Cardiovascular system (hypertensive vascular disease)

- Left ventricular hypertrophy, hypertrophic cardiomyopathy
- Congestive heart failure
- Coronary artery disease and myocardial infarction
- Atrial fibrillation
- Aortic aneurysm
- Aortic dissection
- Carotid artery stenosis
- Peripheral artery disease
- Atherosclerosis

Brain

- Stroke

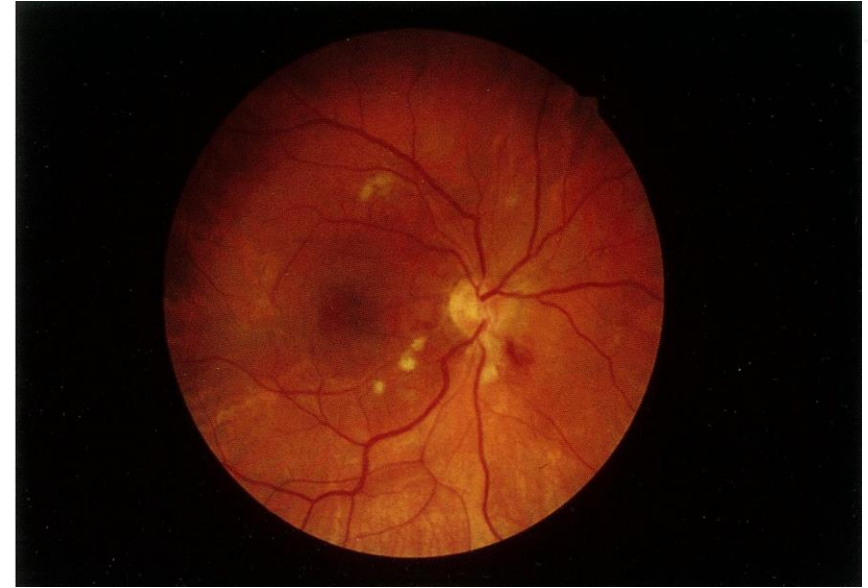
Kidneys

- Hypertensive nephrosclerosis: a renal vascular injury secondary to long-standing arterial hypertension
- Chronic kidney disease

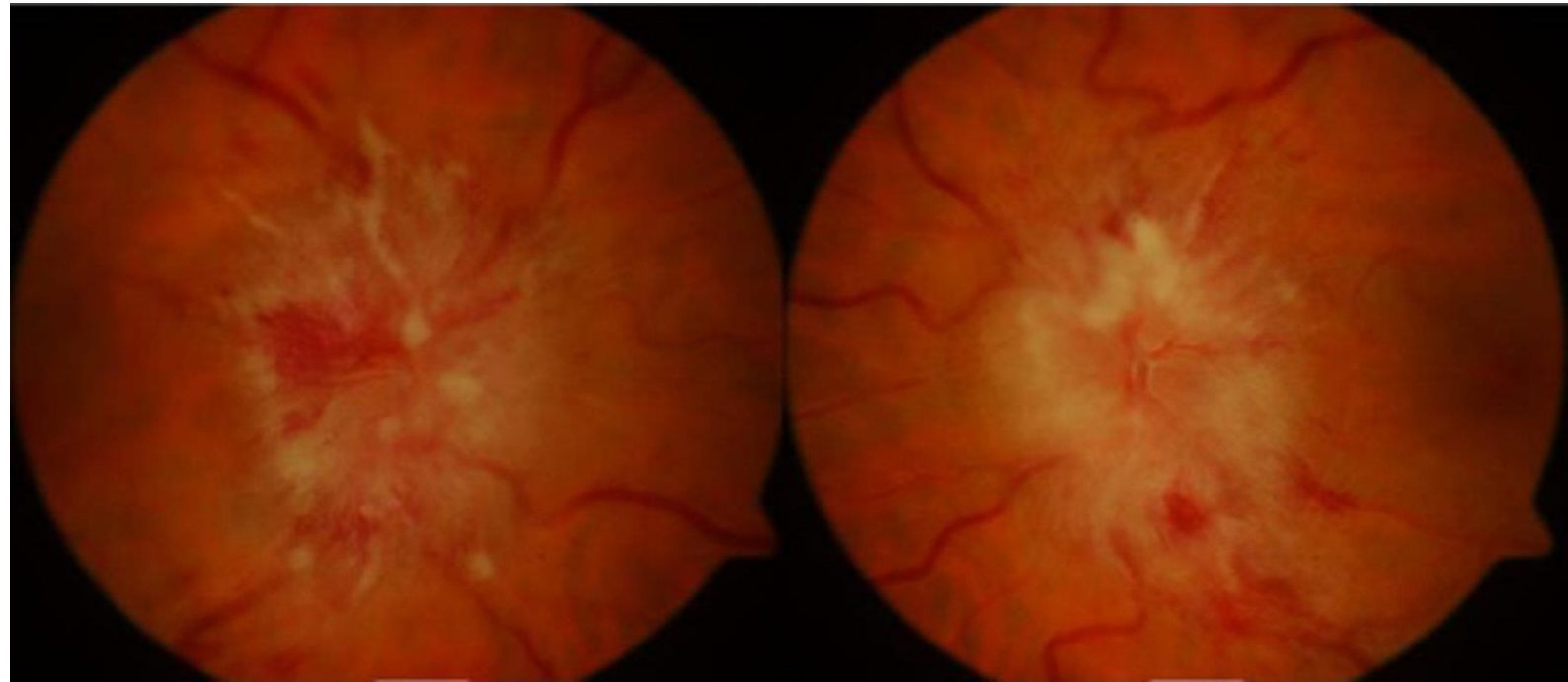
Complications

Eyes

- Hypertensive retinopathy
- Arteriosclerotic and hypertension-related changes of the retinal vessels



The presence of papilledema in a hypertensive patient may indicate a **hypertensive crisis** and warrants urgent lowering of blood pressure.



Hypertensive crises (Hypertensive crisis, Acute severe hypertension)

Preferred terminology

Hypertensive crisis

(acute severe hypertension): systolic blood pressure ≥ 180 mm Hg and/or diastolic blood pressure ≥ 120 mm Hg

Hypertensive urgency:

hypertensive crisis that is either asymptomatic or associated with isolated nonspecific symptoms (e.g., headache, dizziness, or epistaxis) **without signs of acute organ damage**

Hypertensive emergency:

hypertensive crisis **with signs of acute end-organ damage**, mainly in the cardiovascular, central nervous, and renal systems

- **Drug-related**
 - Nonadherence to antihypertensives
 - Drugs that may exacerbate hypertension (e.g., MAO inhibitors, TCAs, NSAIDs, cocaine, amphetamines, ecstasy, stimulant diet pills)
 - Consumption of foods rich in tyramine (e.g., wine, chocolate, aged cheese, cured meat) during therapeutic use of MAOIs
 - Combination or overlap (e.g., due to incorrect washout periods) of MAOIs with other drugs that increase the monoamine concentration (e.g., SSRIs)
- **Pheochromocytoma**
- **Hyperthyroidism**
- **Acute and rapidly progressive renal disorders**
- **Collagen vascular diseases (e.g., SLE)**
- **Eclampsia/Pre-eclampsia**

Clinical features

Clinical features of hypertensive urgency

Asymptomatic or isolated, nonspecific symptoms (e.g., headache, dizziness, or epistaxis)

Clinical features of hypertensive emergency

Cardiac

Heart failure exacerbation, Myocardial infarction, Aortic dissection

Neurological

Hypertensive encephalopathy, stroke

Renal

Acute hypertensive nephrosclerosis, Acute kidney injury

Ophthalmic

Acute hypertensive retinopathy

Other

Microangiopathic hemolytic anemia: fatigue, pallor

Diagnosis and Treatment of Severe Hypertension and Hypertensive Emergency

LEGEND

COR 1

COR 2a

COR 2b

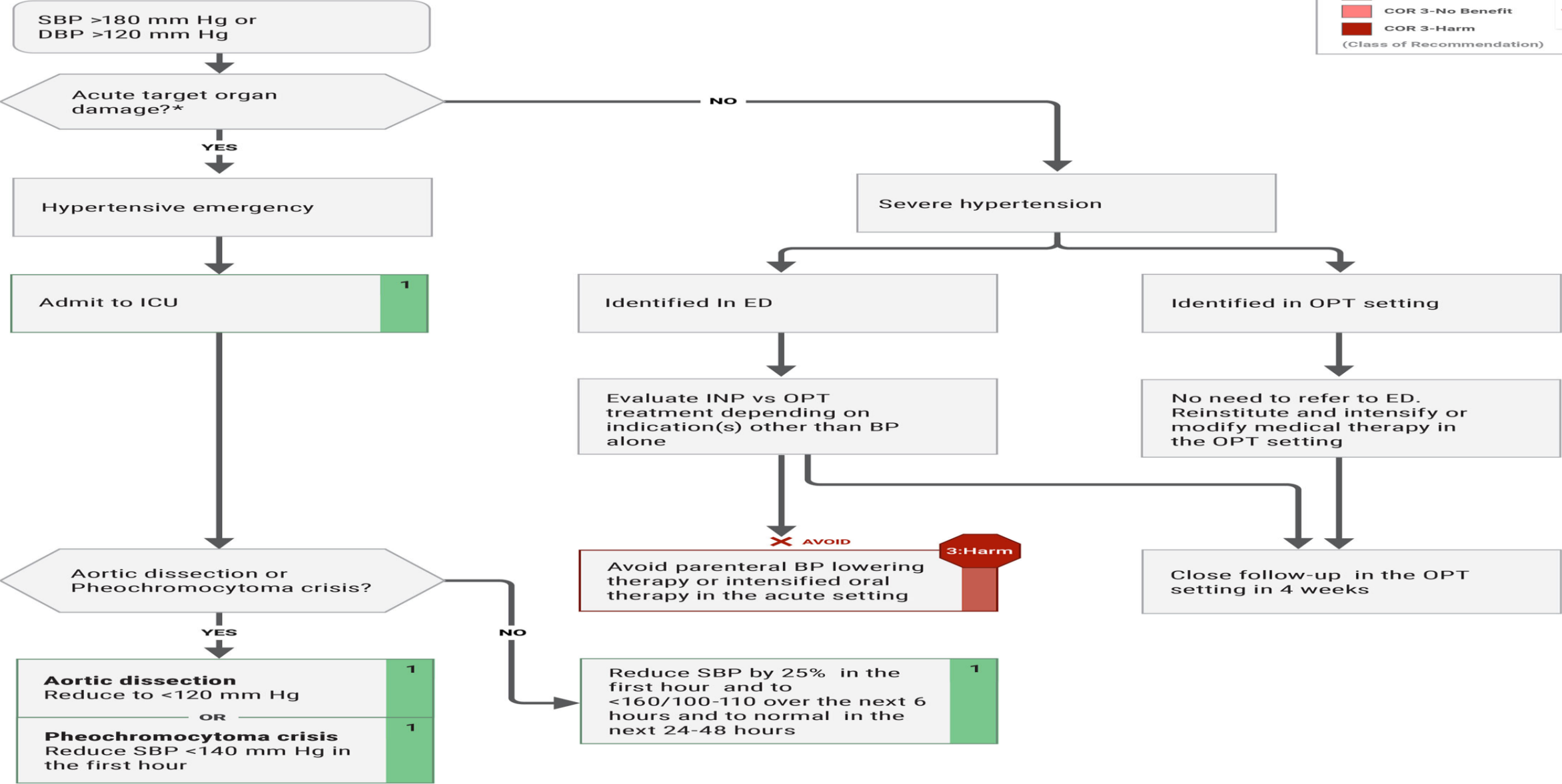
COR 3-No Benefit

COR 3-Harm

(Class of Recommendation)



BMC
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Hypertensive pregnancy disorders

Gestational hypertension

Pregnancy-induced hypertension (defined as systolic blood pressure ≥ 140 mmHg **or** diastolic blood pressure ≥ 90 mmHg on **two** separate measurements at least **4** hours apart) **without** proteinuria **or** end-organ dysfunction.

Diagnosed after **20** weeks' gestation

Does not persist longer than **12** weeks postpartum.

Chronic hypertension

Hypertension diagnosed before pregnancy or in the first 20 weeks of pregnancy

Hypertensive pregnancy disorders

Hypertensive crisis during pregnancy

Systolic blood pressure > 160 mm Hg or diastolic pressure > 110 mm Hg that persists for ≥ 15 min

Preeclampsia

New-onset gestational hypertension **with** proteinuria **or** end-organ dysfunction

Superimposed preeclampsia

Preeclampsia that occurs in a patient with chronic hypertension

Hypertensive pregnancy disorders

HELLP syndrome

A life-threatening form of preeclampsia

Hemolysis, Elevated Liver enzymes, and Low Platelets

May occur without hypertension or proteinuria

Occurrence of new-onset hypertension, proteinuria, or end-organ dysfunction at < 20 weeks' gestation is suggestive of gestational trophoblastic disease.

Eclampsia

New-onset seizures in the absence of other causes ; a convulsive manifestation of hypertensive pregnancy disorders

Postpartum hypertension

Hypertension that persists after delivery, Generally resolves within 12 weeks.

If hypertension lasts > 12 weeks postpartum, a secondary cause should be considered.

Management of urgent hypertensive

Approach

- Initiate antihypertensives for urgent blood pressure control in pregnancy.
- Administer magnesium sulfate for seizure prophylaxis.
- Assess for indications for immediate delivery regardless of gestational age.
 - If present: urgent delivery after maternal hemodynamic stabilization
 - If absent:
 - ✓ ≥ 34 weeks' gestation: Deliver.
 - ✓ 24–34 weeks' gestation: Administer corticosteroids for fetal lung maturity followed by expedited delivery.

Delivery is the only cure for preeclampsia, eclampsia, and HELLP syndrome.

Indications for expedited delivery in hypertensive pregnancy disorders

- Eclampsia (cerebral symptoms)
- Pulmonary edema
- Disseminated intravascular coagulation
- Placental abruption
- Labor or premature rupture of membranes
- Severe oligohydramnios

Hypertensive pregnancy disorders

Eclampsia medical management

- Treat eclamptic seizures.
- Place patient in the left lateral decubitus position
- Start anticonvulsive therapy.
 - First line: magnesium sulfate
- Start antihypertensives for urgent blood pressure control in pregnancy.
- Eclampsia is an indication for immediate delivery regardless of gestational age.
- Delivery should occur only after the mother is stable and seizures have stopped.

References & further reading

- Cian P. McCarthy, John W. McEvoy, Jennifer L. Cluett, Implementing the 2025 Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults: Leveraging Evidence to Push the Boundaries of Clinical Care, Hypertension, (2025). <https://doi.org/10.1161/HYPERTENSIONAHA.125.25418>
- Whelton, PK, Carey, RM et al. 2017 ACC/AHA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults. Hypertension. 2017; 71(6): p.e13–e115. doi: 10.1161/hyp.0000000000000065
- Poon LC, Magee LA, Verlohren S, et al. A literature review and best practice advice for second and third trimester risk stratification, monitoring, and management of pre-eclampsia. International Journal of Gynecology & Obstetrics. 2021; 154(S1): p.3-31. doi: 10.1002/ijgo.13763
- AMBOSS
- Harrison's Principles of Internal Medicine 20th edition ISBN-13: 978-1259644030

Thank You